

and Tegmark 2023; Li et al. 2023). This makes layers a natural mediator for exploratory interventions where using more fine-grained mediators is infeasible, as in Conmy et al. (2023), or where broad characterizations of information flow are sufficient, as in Geva et al. (2023); Sharma, Atkinson, and Bau (2024).

Full layers are rare mediators in mechanistic interpretability. This is because $\mathbf{h}^\ell$ is a **causal bottleneck**, such that all information in the model must pass through it. For example, if we run input x_i through a model, and intervene on $\mathbf{h}^\ell$ to set it to what it would have been given another input x'_i , the model's output would be identical to the case where we simply change the input to x'_i and perform no interventions. That said, interventions to full layers were employed in a pruning study where the motivation was not interpretability (Sajjad et al. 2023); we believe that a more refined intervention technique has the potential to inform our understanding of which model *regions* (as opposed to components) are more responsible for certain behaviors (e.g. Lad, Gurnee, and Tegmark 2024).¹⁸ Past work has used coarse-grained methods based on full layers to investigate factual recall in language models (Geva et al. 2023), and to update these factual associations (Meng et al. 2022, 2023).

The primary advantage of using full layers and submodules as mediators is their small quantity and broad scope of information (high generality). This means that even slow or resource-intensive methods will generally be easy to apply to all layers. In some cases, this is sufficient. However, an obvious disadvantage is that full layers are generally difficult to understand, as they are not particularly sparse nor selective: even if we know that a concept is encoded in a given layer, it is unclear precisely how this information is encoded, composed, or used, and how we might intervene on it without affecting other concepts (Conmy et al. 2023). Thus, layers and submodules will generally be too coarse to explain model behavior or verify mechanistic hypotheses; under these goals, one is better off using full layers as tools to narrow the search space for harder-to-localize mediators (e.g., Brinkmann et al. 2024; Geva et al. 2023). However, if one's goal is localization or model editing, then full layers may be sufficient (Meng et al. 2022, 2023; Sharma, Atkinson, and Bau 2024; Gandikota et al. 2023, 2024).

5.2 Basis-aligned subspaces

Neurons. Compared to a full layer or submodule $\mathbf{h}^\ell$, a neuron $\mathbf{h}_i^\ell$ represents a more fine-grained component that could feasibly represent an individual concept (though we discuss below that this is not often the case due to polysemanticity). A neuron can be considered the smallest meaningful unit in the computation graph $\mathcal{C}$; the neuron's activation h_i^ℓ is a scalar corresponding to a single dimension (1-dimensional subspace) of a hidden representation vector. Each neuron can sometimes be distinguished from others based on its functional role in the network; for instance, Bau et al. (2019a) locate neurons in a machine translation model responsible for detecting or generating items of a particular tense, gender, or number, and causally verify their roles by intervening on their activations in a targeted manner. Bau et al. (2020) locate neurons in a GAN responsible for generating specific types of objects in images, such as trees or windows, and verify this causally by ablating or artificially activating them.

¹⁸ Relatedly, one could intervene on submodules, such as $\mathbf{h}^{\ell\text{-MLP}}$, and observe how this impacts the model, or the features present in the final layer output $\mathbf{h}^\ell$.

Neurons are a natural choice of mediator, as they are both fine-grained (sparse) and easy to exhaustively iterate over (see §6.1); there are $O(\ell \cdot d)$ of them, where d is size of the activation vector $|\mathbf{h}^\ell|$.¹⁹

However, a major disadvantage of neuron-based interpretability methods is *polysemanticity*. Individual neurons are often polysemantic—i.e. they respond to multiple seemingly unrelated concepts simultaneously (Arora et al. 2018), such that a neuron may be *sparse* but not *selective* nor *human-understandable*. This gives them relatively low utility for explaining model behaviors, but they can be useful for verifying mechanistic hypotheses (Geiger et al. 2021) or localizing model behaviors (Vig et al. 2020). For example, if the same neuron were sensitive to capitalized words, animal names, one-digit numbers, among other phenomena, and a researcher were to inspect the activations of that neuron, it would be difficult to disentangle each of these individual roles. Elhage et al. (2022b) investigate polysemanticity and suggest that neural networks represent features through linear superposition, where they represent features along non-basis-aligned linear subspaces, resulting in interpretable units being smeared across multiple neurons. In other words, in an activation vector $\mathbf{h}^\ell$ of size $|\mathbf{h}^\ell| = d$, a model can encode $k \gg d$ concepts as directions (Park, Choe, and Veitch 2023), such that only a sparse subset of concepts are active given a particular input.

Basis-aligned multidimensional subspaces. The computations of a neuron $\mathbf{h}_i^\ell$ are often not independent: *sets* of neurons can compose to encode some concept. For example, in language models, subsets of neurons $\{\mathbf{h}_i^\ell, \mathbf{h}_j^\ell, \dots\} \in \mathbf{h}^\ell$ can be implicated in encoding gender bias (Vig et al. 2020), and implementing latent linguistic phenomena (Finlayson et al. 2021; Mueller, Xia, and Linzen 2022; Bau et al. 2019a; Lakretz et al. 2019). Thus, some early mechanistic interpretability work employed heuristic-based searches over sets of neuron responsible for some behavior (e.g., Bau et al. 2019b; Vig et al. 2020; Cao, Sanh, and Rush 2021; Antverg and Belinkov 2022). This is a generalization of individual neurons as mediators, where multiple dimensions in activation space are intervened upon simultaneously.

Sets of neurons have strictly more expressive power than individual neurons, and thus have the potential to explain model behavior more broadly than finer-grained mediators. In other words, one can conceptualize neuron groups as trading off some sparsity for increased generality. If a concept is encoded across multiple neurons, then neuron groups may also enable more human-interpretable interventions than one-dimensional subspaces. Despite this, basis-aligned multidimensional subspaces are not commonly studied. This is for two main reasons: (1) There is a combinatorial explosion when we are allowed to search over arbitrarily-sized sets of neurons; if using exhaustive search, this increases the number of required forward passes from $O(\ell \cdot d)$ to $O(2^{\ell \cdot d})$, which makes this intractable. (2) Furthermore, interpretable concepts are not guaranteed to be aligned to neuron bases, meaning that groups of neurons still do not directly address the problem of polysemanticity (Morcos et al. 2018; Chughtai, Chan, and Nanda 2023; Wang et al. 2023).

¹⁹ We use big- O notation because the exact number will depend on whether one includes just the activation vectors at the end of a layer, or additionally includes the vectors output by the MLP and/or attention blocks, (among other possible vectors). Also, if looking at intermediate MLP neurons, the number is typically $e \cdot |\mathbf{h}^\ell|$, where e is some constant expansion factor, typically in $[1.5, 4.0]$.

Attention heads. Similar to neurons, attention heads are fundamental components of Transformer-based neural networks. They mediate the flow of information between token positions (Vaswani et al. 2017); thus, using attention heads as units of causal analysis can help us understand how models synthesize contextual information (Ma et al. 2021; Neo, Cohen, and Barez 2024) to predict subsequent tokens (Wang et al. 2023; Hanna, Liu, and Variengien 2023b; Prakash et al. 2024; García-Carrasco, Maté, and Trujillo 2024; Brinkmann et al. 2024).

Each head A_i^ℓ in layer ℓ can be understood as an independent operation contributing a vector output $\mathbf{a}_i^\ell$; the outputs of all heads are concatenated and projected to form the output of the attention block $\mathbf{h}^{\ell\text{-Attn}}$, which is then added into the residual stream $\mathbf{h}^\ell$.²⁰ For example, some heads specialize on syntactic relationships (Chen et al. 2024a), others on semantic relationships such as co-reference (Vig et al. 2020), and others still on maintaining long-range dependencies in text (Wu et al. 2024a). Attention heads have also been directly implicated in acquiring the ability to perform in-context learning (Olsson et al. 2022; Brown et al. 2020), or to detect and encode functions in latent space (Todd et al. 2024; Feucht et al. 2025).

Attention heads are attractive mediators because they are easily enumerable: there are generally far fewer attention heads $O(\ell \cdot |A^\ell|)$ than neurons $O(\ell \cdot d)$ in a model, as $|A^\ell| \ll d$ in typical Transformer-based models. Attention heads also track multi-token relationships. However, in contrast to the activation h_i^ℓ of a neuron, the output of an attention head $\mathbf{a}_i^\ell$ is multidimensional. Thus, it is difficult to directly interpret the full set of functional roles a single head might have: attention heads are almost always polysemantic, so one cannot typically determine the function(s) of an attention head solely by observing its outputs (Janiak, cmathw, and Heimersheim 2023).²¹ It has additionally been observed that intervening on an attention head can cause other attention heads to compensate, which further complicates causal analyses (Jermyn, Olah, and Henighan 2023; Wang et al. 2023; McGrath et al. 2023).²² To summarize, attention heads are easier to exhaustively search over than (sets of) neurons, but have the same issues of low selectivity and human-interpretability.

5.3 Non-basis-aligned spaces

Non-basis-aligned multidimensional subspaces. Due to their polysemanticity, neurons, attention heads, and sets thereof do not necessarily correspond to cleanly interpretable concepts. For example, individual neurons typically activate on many seemingly unrelated inputs (Elhage et al. 2022b), and this issue cannot be cleanly resolved by adding more dimensions. This is because the features may actually be encoded in directions or

²⁰ This is the *residual stream perspective* (Elhage et al. 2021) of Transformers, which has been adopted in recent interpretability research (Ferrando et al. 2024). The *residual stream perspective* suggests that the residual stream, which comprises the sum of the outputs of all the previous layers and the original input embedding, acts as a passive communication channel through which the MLP and attention submodules route the information they add.

²¹ However, there is initial evidence that some dimensions of an attention head’s output can be meaningfully explained (Merullo, Eickhoff, and Pavlick 2024a,b; Hu et al. 2025). Thus, by decomposing the vector output of a head into smaller subspaces or even individual neurons, it may be easier to explain the set of functional roles of a given head.

²² This phenomenon where downstream components only have causal relevance after an upstream component has been removed is sometimes called **preemption** in the causality literature (Mueller 2024), or the “Hydra effect” in mechanistic interpretability (McGrath et al. 2023). Preemption is not limited to attention heads; future work should analyze how common preemption is between other types of components, such as MLP submodules.